PA194

A 194-state binary Turing machine for the consistency of Peano arithmetic

Construction and verification report

13 November 2026

Source snapshot: `5be8b9236df6a6d308195e62a906a173dc70d00a`

Abstract

We construct an explicit deterministic single-tape, two-symbol Turing machine with 194 working states and give an ordinary mathematical proof that PA proves its blank-tape nonhalting equivalent to Con(PA). The construction uses the existing 190-item separate-vacuity axiom emitter. A new binary association admits eight simultaneously valid reader identifications: 134 inherited controls plus 68 axiom-reader controls, less eight, gives 194. This improves the supplied 195-state construction by one state and the earlier separate-emitter implementation by two.

The package contains the complete 388-transition table, deterministic reconstruction, the 3,673-site source program, 32 exhaustive local simulation cases, an exact 58,154-step blank startup, and all 4,780 symbolic primitive-branch checks for both the unfolded and folded tables. An independent checker reconstructs the machine and source graph without using the candidate builder. The comparison covers all 31 incoming archives at the pinned revision and the direct PA research directory. The consistency equivalence is an ordinary mathematical proof, with explicit proof-calculus and ten-entry command interfaces; a new end-to-end Lean or Rocq kernel theorem is not claimed.

194 working states 2 symbols 388 transitions

$$\mathrm{PA} \vdash \neg\,\mathrm{Halt}(\mathrm{PA194}, \mathrm{blank}) \;\leftrightarrow\; \mathrm{Con}(\mathrm{PA}).$$

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1 The result and its exact conventions

The definitive file is `pa194/pa194.tm`. It uses working states $0, \ldots, 193$, initial state 0, and the separate halt label 194. The tape alphabet is $\{0, 1\}$, the tape extends infinitely in both directions, every initial cell contains zero, and the initial head coordinate is zero. There are exactly two outgoing transitions from each working state, one for each read symbol. Every transition moves one cell left or right; there are no stationary moves. The halt label has no outgoing transition and is excluded from the working-state count.

A row $q\ b\ w\ D\ q'$ means: in state q, reading b, write w, move in direction $D \in \{L, R\}$, and enter q'. Appendix A prints every transition. The SHA-256 of the exact five-column file, including its final newline, is

`9087d70fc0e1878c38ceff928e6e7e0999c86b6efed00a5ec353821c00641c87`.

Quantity	Verified value
Working states / rows	194 / 388
Initial state / halt label	0 / 194
All working states graph-reachable	Yes
First transition on blank	$0, 0 \mapsto 1, R, 1$
Unique halting transition	$63, 1 \mapsto 1, R, 194$
Source-dispatch root in dense labels	75
Source primitive occurrences	$3,673$
Symbolic primitive branches	$4,780$
Longest source address	38 bits
Exact blank startup	$58,154$ literal transitions

Graph reachability is a finite property of the transition graph. It is not an assertion that every state is reached on the actual blank run. The internal theorem concerns the arithmetic sentence for this exact table. It does not assume that PA is consistent; actual nonhalting follows if PA is consistent.

1.1 What is new

The separate-vacuity emitter and its consistency calibration were already proposed in the supplied research [2, 1]. This work does not claim their invention. The new result is an association and allocation that realizes that emitter in 194 states, with eight compatible physical reader identifications and complete source, runtime, and blank-start evidence.

The supplied PA195 uses a 191-item emitter and allocates $134 + 68 - 7 = 195$ controls. The earlier separate196 uses a 190-item emitter and allocates $134 + 68 - 6 = 196$. Our new association retains 68 raw readers and validates eight folds:

$$\boxed{134 + 68 - 8 = 194.}$$

The fixed part has 36 tape/register-engine controls, 13 primitive routers, and 85 readers outside this AX allocation. The final reader count is $85 + 60 = 145$, giving $36 + 13 + 145 = 194$. The state saving concerns the actual compiled table, rather than source length alone.

1.2 Proof and formalization status

There are three different claims in this package. First, the literal table and every finite certificate can be reconstructed and checked by the supplied programs. Second, the arguments below prove the associated

halting and consistency equivalences in ordinary mathematical form and explain why the transformations are formalizable in PA. Third, the inherited formal development supplies many of the arithmetic, source, and register-engine facts used by that argument.

These claims should not be conflated. The new 194-state endpoint has not been elaborated as a fresh Lean or Rocq theorem. In particular, the published PA196 internal implication from nonhalting to consistency does not by itself supply the reverse implication. The written reverse proof in the PA195 package uses syntactic PA proof transformations; our calibration preserves that distinction [1, 3].

2 Comparison of the supplied designs

The comparison fixes ProveIt commit `5be8b9236df6a6d308195e62a906a173dc70d00a`. The exact user-designated PA195 ZIP matches Git blob `deb9187261c34ae6a3876a9bfff773ff31737415`. The current incoming directory contains 31 ZIP archives, totaling 233,712,151 bytes. The fresh recursive census visits 198 ZIP occurrences representing 44 distinct archive payloads and also scans the direct PA machine files. It recognizes complete numeric five-column binary tables in `.tm` and `.txt` members. It does not purport to recognize every machine embedded in arbitrary source code or JSON.

That census finds 2,719 literal table occurrences, 576 distinct byte strings, and 566 labeled transition functions. These are inventory counts: they include duplicate ancestors, experiments, rejected allocations, and small controls. Among the incoming tables, 17 195-state occurrences form seven rooted action-preserving graph classes: the supported PA195 and six earlier rejected or uncertified designs. No recognized incoming table has between seven and 194 working states. This is a statement about the pinned corpus, not a lower bound on possible machines. Full hashes, member locators, and parser scope are recorded in `comparison/corpus/census_current.json`.

Design	States	AX items	Source sites	Branches
Published PA196	196	201	3,812	4,959
Prior diagonal196	196	191	3,700	4,815
Prior separate196	196	190	3,673	4,780
Supplied PA195	195	191	3,700	4,815
New alternative diagonal layout	195	191	3,700	4,815
PA194	**194**	**190**	**3,673**	**4,780**

The prior three central rows and their proof status are documented in the inherited comparison [1, 2]. The new alternative 195-state layout uses 67 raw AX readers and six folds. It was a useful search seed, but is not the requested improvement. The delivered 194-state layout uses the exact separate190 macro word, default head order, original formula parameters, and original command operations. It does not use the later H-only Shift or MP-return-in-H commands.

2.1 The larger families and the additional incoming reports

The U208 family contributes finite history-dependent reader reductions and several equivalent presentations. The previous audit distinguishes 20 designated U208 graph classes among 27 presentations. Their principal one-state saving identifies particular old native roles that have already disappeared from PA196. It cannot simply be subtracted again from PA195. James Chang's historical PA209 is also distinct from the U209 ancestors inside those archives [1, 3].

The earlier PA195 audit retains 25 supported 196-state graph classes, including alternate axiom orders, source rotations, contextual closing instructions, a register swap, H-only Shift, combined commands,

diagonal/separate emitters, and invariant-based zero jumps. Its full 196-state census also records many partially checked or unsuccessful allocations. We preserve that classified ledger and its exact normalized tables in the unchanged reference packet. This avoids promoting a small-looking file to an established consistency machine merely because of its advertised state count.

Four archives extend the older 27-archive snapshot. PA195 itself supplies the new baseline. The cross-archive report combines diagonal/separate emitters with H-only Shift and successful MP in H; its selected implementations remain at 196 states, despite reducing the source to 3,676 or 3,649 sites. The later reduction audit supplies additional 196-state MP compositions and 197-state short-negation compositions. The later research report proves an 18-instruction negation macro and a clean-scratch invariant, but its complete separate-emitter table has 197 states, 3,441 sites, and 4,468 contracts [4, 5, 6]. These examples make the practical issue explicit: shortening a source routine can remove a helper on which physical sharing depends.

Appendix B gives a contribution/status entry for all 31 archives. The direct PA directory, ZFC directory, and ConciseIPRA sources are treated as additional reference material, not as extra incoming ZIPs.

3 The exact source and the eight-state quotient

Let R be `pa194/unfolded202.native.tm`. Its designated native start is 74 and halt is 254. The builder reads the saved association `pa194/grammar.json`, retains the 134 fixed controls, and creates 68 AX readers. It then constructs the folded table M using the following disjoint identifications. Each left-hand control is removed.

$$2019 \mapsto 2025, \quad 2024 \mapsto 2005, \quad 2031 \mapsto 2052, \quad 2034 \mapsto 2020,$$
$$2042 \mapsto 2029, \quad 2055 \mapsto 2053, \quad 2058 \mapsto 2059, \quad 2061 \mapsto 2038.$$

All replacement rows and target redirects are explicit in `pa194/folds.json`. The native table is then relabeled bijectively to the dense 194-state presentation. Relabeling neither changes the tape convention nor identifies any further states.

3.1 Preserving the source program

Reassociation changes binary addresses on the tape, so preservation of the flattened macro word alone is insufficient. The builder expands the complete source, computes its primitive instruction at every ordinal, and resolves every successor. The resulting 3,673-site list contains 1,457 increments, 1,107 conditional decrements, 1,108 jumps, and one halt. Each decrement contributes two contracts, so the total is $3,673 + 1,107 = 4,780$.

The ordered primitive word and every ordinal successor are exactly equal to the supplied separate196 source. Two independently structured resolvers agree. The source seed remains ordinal 25, at address `100000`; it is a decrement of register 8, whose zero successor is 27 and positive successor is 26. No source-level semantic optimization is hidden in this new allocation.

The emitter comparison is also exact. Its 190 macro items equal the prior separate190 word. A symbolic free-algebra execution checks the entire output stack and final formula, with independent symbols for all input parameters. This is an identity of expression trees, rather than agreement on finitely many selected numeric formula codes.

3.2 The predecessor-bit invariant

For each pair $a \mapsto b$, every incoming transition to a in R writes zero and moves right; every incoming transition to b writes one and moves right. The complete census has 23 incoming rows. Neither endpoint is the initial control. Thus every reachable parent configuration satisfies

$$q = a \Longrightarrow t(h-1) = 0, \qquad q = b \Longrightarrow t(h-1) = 1.$$

Here t is the tape and h the current head coordinate. This is a one-step induction on the parent run. The invariant records exactly the information needed to distinguish two old controls after they have been identified physically.

3.3 All 32 local cases

For each of the 16 old controls and each current read bit, the checker starts the folded machine with the prescribed bit at coordinate -1 and the arbitrary current bit at coordinate 0. It compares the result with one transition of R, after applying the control quotient. Every read and write before the endpoint lies at -1 or 0; both cells must be restored to exactly the parent endpoint contents. The final head position and target control must also agree.

The 32 successful blocks have lengths

$$12 \text{ of length } 1, \quad 16 \text{ of length } 3, \quad 3 \text{ of length } 5, \quad 1 \text{ of length } 7.$$

Since no other tape cell is inspected, each check is valid for every extension of those two cells to the rest of the tape. Every unaffected parent row commutes with the quotient in one step. All checks use the *final simultaneous table*: an identification is not certified against a helper that another identification later changes.

The unique seven-step case starts from raw state 2034, reading zero, with predecessor bit zero. Its parent transition is $0R166$. The following folded trace touches only the two zero cells, restores both, and ends at head coordinate one in state 166:

Step	Control	Head	Read	Action
0	2020	0	0	0L2054
1	2054	-1	0	0R2053
2	2053	0	0	0L2057
3	2057	-1	0	0R2059
4	2059	0	0	0L2021
5	2021	-1	0	0R2022
6	2022	0	0	0R166
7	166	1	—	Parent endpoint

Theorem 3.1 (Halting preservation by the quotient)**.** *PA proves that the blank-start unfolded 202-state machine halts if and only if PA194 halts.*

Proof. Map (q, h, t) to $(\sigma(q), h, t)$, where σ is the eight-pair control quotient and fixes the start and halt. The invariant and the 32 finite identities simulate each parent step by between one and seven folded steps. Concatenation therefore transforms any parent halting trace into a folded halting trace, preserving the full tape at comparison boundaries.

Conversely, suppose the folded machine halts after T steps. If the parent had not halted within its first T steps, its first $T + 1$ steps would yield more than T defined folded steps. Determinism and the

absence of outgoing rows at halt give a contradiction. An intermediate premature halt is excluded by each checked local block.

The invariant is induction on a coded finite trace; the local checks are fixed finite cases; concatenation and the $T + 1$ bounded iteration are primitive recursive. Their correctness is proved in PA. No consistency or nonhalting assumption is used in this quotient theorem. □

4 Runtime, startup, and independent verification

The source-to-tape checker represents arbitrary natural register values, a symbolic affine gap, and an arbitrary register-list tail. It reduces the unchanged register excursions using the inherited list/register loop lemmas and checks complete successor frames. It checks both branches of every conditional decrement and checks the halt leaf explicitly. The relevant protected register-engine rows are unchanged from the supplied PA195 and PA196 tables.

All 4,780 symbolic contracts succeed for R and for PA194. The longest symbolic path is 506 steps in each case. Total symbolic paths have 1,546,047 steps before folding and 1,571,586 after folding. These numbers measure finite symbolic certificates. They are not uniform literal time bounds for unary register operations on arbitrarily large natural values.

Starting at dense control 0 on the all-zero tape, an exact literal simulation reaches native control 84 (dense 75), head coordinate -2, after 58,154 steps. It has the complete standard source seed: header gap 48 and physical register file

$$(0, 0, 0, 0, 0, 0, 2, 0, 0, \underbrace{1, \ldots, 1}_{26 \text{ entries}}).$$

The certificate lists all 66 one positions and verifies that every other tape cell remains zero. It verifies an exact frame, rather than merely observing the dispatcher state or a small tape window.

The second verifier reconstructs raw202 from the saved grammar and pinned reference rows, independently rebuilds the quotient, checks dense/native conjugacy, counts all 194 graph-reachable controls, and replays the 32 two-cell cases. It independently expands and resolves the source against separate190, then generates its own contract list. The symbolic execution engine and its boxed register lemmas are still shared, explicitly identified dependencies. Independence of the reconstruction and contract generation should not be misdescribed as two independently formalized theorem provers.

The verification receipts are `pa194/validation.json` and the files in `verification/`. Each clearly distinguishes executable certificates from a new proof-assistant kernel theorem.

5 The PA-provable consistency equivalence

Let M_{194} be the exact table `pa194/pa194.tm`, started in state zero on the all-zero tape, and let H_{194} assert that this configuration has a finite halting trace. The halt label is 194; it is not one of the 194 working states. Fix a conventional recursively checked first-order PA proof predicate $\mathrm{Prf}_{\mathrm{PA}}$, and put

$$\mathrm{Con(PA)} \equiv \neg\exists p \, \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner).$$

Theorem 5.1. *The delivered table satisfies*

$$\mathrm{PA} \vdash (\neg H_{194} \leftrightarrow \mathrm{Con(PA)}).$$

This is an ordinary mathematical proof about PA-provability. The finite certificates in the package verify its candidate-specific program facts; the new endpoint has not been exported as a Lean or Rocq kernel theorem.

The change in logical source is the separate-diagonal emitter already present in the supplied separate196 design. Its new association and eight compatible physical folds give the present state count. We give the changed command compiler explicitly, since the old nine-entry compiler cannot be reused with its old indices. The commands below are the original commands, including the test $t + h = 1$; they are not the later H-only Shift or MP-in-H variants.

5.1 Exact numerical calculus and emitter

Use Cantor pairing and the four constructors

$$p(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad I(A,B) = 4p(A,B),$$

$$M(a,b) = 4p(a,b) + 1, \quad U(a,B) = 4p(B,a) + 2, \quad E(a,b) = 4p(a,b) + 3.$$

Thus U deliberately transposes its arguments. Write $\neg A = I(A,1)$, $A \wedge B = \neg(A \to \neg B)$, and $\exists x_a A = \neg U(a, \neg A)$. Code 0 is truth, while $1 = M(0,0)$ is the designated contradiction. In particular, $I(0,0) = 0$ is an actual numerical alias.

The six retained axiom families, in their original push order, are

$$\begin{aligned}
h_1(D,H) &= (\exists x_D\, U(D,H)) \to H,\\
h_2(D,A,H) &= U(D,U(A,H)) \to U(A,U(D,H)),\\
h_3(D,A,H) &= M(H,D) \to (E(H,A) \to M(A,D)),\\
h_4(D,A,H) &= E(H,D) \to (M(A,H) \to M(A,D)),\\
h_5(D,H) &= \exists x_D E(H,D),\\
h_6(D,A,H) &= U(D,A) \to (U(D,A \to H) \to U(D,H)).
\end{aligned}$$

Also retain the dominated propositional family

$$\rho_8(D,A,H,u) = ((H \to u) \to D) \to ((D \to H) \to (A \to H)), \qquad u \le D,$$

modus ponens, and generalization $B \mapsto U(i,B)$ on closed derivations. Here "closed" means without undischarged hypotheses; the formula itself may contain free variable labels.

Put

$$\begin{aligned}
\Gamma(B) &= B \to U(B,B),\\
V(a,b,c) &= M(a,b) \to U(c,M(a,b)),\\
P(a,b,c) &= c \to M(c,b),\\
S(a,b,c) &= U\big(b,(M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big),\\
Q(a,b,c) &= U(c,S(a,b,c) \to \neg P(a,b,c)),\\
G(a,b,c) &= \neg(V(a,b,c) \to Q(a,b,c)).
\end{aligned}$$

The separate calculus $\mathcal{C}_s$ adds every $\Gamma(B)$ and every $G(a,b,c)$ with $c \neq a,b$. A failed guard emits 0. This adds no unlisted logical rule or arithmetic axiom. Truth is derivable already: $\rho_8(0,0,0,0) = 0$.

For an incoming store (t,w,d,h), command $\mathsf{AX}(c,x,y)$ sets $D = d + x$, $H = h + y$, $A = c$, and pushes, in order,

$$t,\ U(d,h),\ \rho_8(D,A,H,d),\ h_1,\ h_2,\ h_3,\ h_4,\ h_5,\ h_6,\ \Gamma(A).$$

It leaves $G(H,D,A)$ in the top register if $A \neq H,D$, and otherwise leaves 0; both caches become zero. The list displayed is push order, so its last entry is at the top of the resulting stack. The complete

ten-entry stack and final top are proved as identities in a free term algebra, with all six incoming parameters symbolic, by `mathematics/check_emitter_and_words.py`. Its grammar expansion equals the archived separate196 word, and equals the PA195 word with the `cons` at zero-based index 184 deleted. This is a 190-token word. The final guard and cleanup blocks are unchanged.

5.2 Proof equivalence with the calibrated diagonal calculus

The PA195 calculus $\mathcal{C}_\Gamma$ has the same six heads, ρ_8, and rules, but its guarded axiom is

$$F_\Gamma(a,b,c) = \neg(\Gamma(c) \to (V(a,b,c) \to Q(a,b,c))).$$

The retained propositional basis and h_1 already derive the classical implication and negation templates, independently of either gate. In particular the restricted ρ_8 supplies the usual positive basis and Peirce law, and h_1 supplies explosion.

Lemma 5.2. *There are primitive-recursive, conclusion-preserving proof translations in both directions between $\mathcal{C}_s$ and $\mathcal{C}_\Gamma$, whose correctness PA proves.*

Proof. Classical propositional templates extract $\Gamma(c)$ and $\neg(V \to Q) = G$ from F_Γ. To supply an arbitrary $\Gamma(B)$, use the guarded instance with $c = B$ and $a = b = B + 1$; its guard always succeeds. Conversely, proofs of $\Gamma(c)$ and $\neg(V \to Q)$ give $\neg(\Gamma(c) \to (V \to Q))$ by a fixed propositional template. All other proof constructors are copied. These transformations recurse on the finite derivation and use only code formation, concatenation, and the displayed parameter choice. PA verifies the finitely many templates and proves correctness by induction on that recursion. No substitution into a freshness-restricted axiom is used. □

For completeness, we specify the logical calibration being inherited. The preserved PA195 report (`reference/pa195_conpa`), in its section `pa_calibration.tex`, constructs PA-verifiable proof translations giving

$$\mathrm{PA} \vdash (\exists p\ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner) \leftrightarrow \exists d\ \mathrm{Prf}_\Gamma(d, 1)). \tag{1}$$

These are syntactic transformations, not an inference from the existence of a standard model. In the forward direction the published PA196 refutation compiler uses equality through direct and reverse membership rewriting, neighborhood equality, and an equality witness. The preserved diagonal proof replaces exactly the reverse and neighborhood suppliers. Its negative-property specialization uses direct rewriting, the retained equality witness, and membership vacuity; it does not assume the equality-vacuity interface being recovered. Choosing a fresh label as the code of the relevant formula gives the required Γ instance. Equality cases of numerical labels are handled explicitly. These replacements are uniform finite proof templates.

The reverse translation interprets $M(a,b)$ by the bit predicate $\mathrm{Bit}(x_a, x_b)$, $E(a,b)$ by $x_a = x_b$, and U by universal quantification. PA proves $\mathrm{Bit}(s,z) \to z < s$ and hence $\neg\,\mathrm{Bit}(z,z)$. Every label in a formula of code B is strictly below B, so $\Gamma(B)$ is a valid vacuity instance. The guarded separation supplier uses the PA proof template

$$\forall n\, \exists s < 2^n\, \forall z < n\, (\mathrm{Bit}(s,z) \leftrightarrow \theta(z, \vec{u}))$$

for an arbitrary formula θ: induct on n, adding 2^n exactly in the case $\theta(n, \vec{u})$. This procedure constructs one induction instance syntactically; it does not evaluate θ or assume reflection for PA. When the source labels $a = b$, the membership antecedent under that binder is $\mathrm{Bit}(z,z)$ and is false; the preserved supplier uses the corresponding singleton comprehension case. All substitutions are capture avoiding. The truth alias is handled by the uniform constructor equivalence

$$T(I(A,B)) \leftrightarrow (T(A) \to T(B)),$$

including $A = B = 0$, rather than by an incorrect syntactic identity. False code 1 translates to $\mathrm{Bit}(x_0, x_0)$, whose negation PA proves even under a binder for label zero.

Thus the retained head suppliers, generalization, and modus ponens have uniform PA proof constructors. Induction on a coded derivation checks the translation for every proof code. This proves the reverse of (1) internally, including nonstandard codes in models of PA. The detailed constructions and their upstream source references are preserved verbatim in the cited section. Composing them with Lemma 5.2 gives the same biconditional with $\mathrm{Prf}_s(d, 1)$.

5.3 The exact ten-entry command compiler

Let $[a]w = p(a, w)$ and let hd, tl be Cantor projections. The unchanged commands act by

$$\mathsf{S}(t, w, d, h) = \begin{cases} \text{halt}, & t + h = 1, \\ (h, \mathrm{tl}(w), d, \mathrm{hd}(w)), & t + h \neq 1, \end{cases}$$

and, for $v = d + x$,

$$\mathsf{MP}(x)(t, w, d, h) = \begin{cases} (v, [0]w, v, 0), & I(t, v) = h, \\ (0, w, v, 0), & I(t, v) \neq h. \end{cases}$$

The AX update is the ten-entry update above. Write q_j for its jth entry counted from the stack top. The reversal is explicit:

j	1	2	3	4	5	6	7	8	9	10
q_j	$\Gamma(A)$	h_6	h_5	h_4	h_3	h_2	h_1	ρ_8	$U(d, h)$	t.

The gate and $q_1, \ldots, q_8$ have residue zero modulo four; q_9 has residue two and is at least two. Directly applying Shift therefore proves, without a restriction on the saved top t,

$$\mathsf{AX}(c, x, y)\mathsf{S}^9(t, w, d, h) = (\rho_8(D, c, H, d), [t]w, 0, U(d, h)), \tag{2}$$

$$\mathsf{AX}(c, x, y)\mathsf{S}^{10}(t, w, d, h) = (U(d, h), w, 0, t). \tag{3}$$

Every tested pair before these endpoints has sum different from one.

Define

$$\begin{aligned} \mathrm{pop} &= \mathsf{AX}(0, 0, 0)\mathsf{S}^{11}, \\ \mathrm{push} &= \mathsf{AX}(0, 0, 0)\mathsf{S}^{9}\mathsf{MP}(0), \\ \mathrm{discard} &= \mathsf{MP}(1)\mathsf{AX}(1, 0, 0)\mathsf{S}^{10}\mathsf{MP}(0). \end{aligned}$$

Their exact identities are

$$\begin{aligned} \mathrm{pop}(t, [a]w, d, h) &= (t, w, 0, a), \\ \mathrm{push}(t, w, d, h) &= (0, [t]w, 0, 0), \\ \mathrm{discard}(t, w, 0, 0) &= (0, w, 0, 0). \end{aligned}$$

For pop, append one safe Shift to (3). For push, the MP comparison after (2) compares an implication code with a universal code, so fails by their different residues. For discard, the first MP fails because $I(t, 1) > 0$, setting $d = 1$. Equation (3) then gives $(U(1, 0), w, 0, 0)$; the final MP fails because $I(U(1, 0), 0) > 0$. These identities include $t = 1$.

For $1 \leq j \leq 7$ define

$$\mathrm{select}_j(D, H, A) = \mathrm{discard}\,\mathsf{AX}(A, D, H)\mathsf{S}^{j+1}\mathrm{pop}^{\,9-j}.$$

It returns $(q_j, w, 0, 0)$ from $(t, w, 0, 0)$. After the direct shifts its top is q_j, its cache is q_{j+1}, and its remaining stack consists of $q_{j+2}, \dots, q_{10}$ and the protected old w. Each pop preserves that top and consumes one entry; the final cache is $q_{10} = 0$ because discard saved truth. Thus $\Gamma(B)$ uses $j = 1, D = H = 0, A = B$, while $h_1, \dots, h_6$ use respectively $j = 7, 6, 5, 4, 3, 2$.

Let offset(i) be empty for $i = 0$ and $\mathsf{MP}(i)$ otherwise. From $(0, w, 0, 0)$ it sets $d = i$ and changes nothing else. The dominated head is selected, for $u \le D$, by

$$\text{discard offset}(u)\,\mathsf{AX}(A, D-u, H)\mathsf{S}^9\text{pop}.$$

The identity $u + (D - u) = D$ gives the desired ρ_8; the remaining saved zero is removed by the final pop. The gate is selected by discard $\mathsf{AX}(c, b, a)\text{pop}^{10}$. It returns the guarded $G(a, b, c)$ and preserves w.

Generalization is implemented by

$$\text{gen}_i = \text{push offset}(i)\,\mathsf{S}\,\mathsf{AX}(i, 0, 0)\mathsf{S}^{10}.$$

After push and offset, Shift transfers the saved formula a to h without testing a; equation (3) then returns $(U(i, a), w, 0, 0)$. For modus ponens use

$$\text{suffix}_b = \mathsf{SSMP}(b)\mathsf{AX}(b, 0, 0)\mathsf{S}^{11}.$$

On $(0, [a][I(a, b)]w, 0, 0)$ it halts at the second Shift if $a = 1$; otherwise MP succeeds and yields $(b, [0]w, b, 0)$, after which the AX–Shift word returns $(b, w, 0, 0)$.

Compile a derivation by selecting its axiom, by applying gen_i to a compiled generalization premise, and by

$$C(q)\,\text{push}\,C(p)\,\text{push}\,\text{suffix}_b$$

for premises $p : a$ and $q : I(a, b)$. Induction on the derivation proves that $C(d)$, from every $(t, w, 0, 0)$, either halts or returns $(a, w, 0, 0)$. A refutation therefore supplies the universally halting word

$$W_d = \mathsf{AX}(0, 0, 0)\,C(d)\,\mathsf{S}.$$

The first AX clears both caches from any incoming store; no reset of the live stack or scheduler store is assumed. All identities are equations of primitive-recursive arithmetic functions, with the displayed residue and positivity proofs; PA proves their universal closures and the compiler induction. The independent symbolic word checker also replays these identities, including both MP-suffix cases.

5.4 Scheduler, source binding, and exact physical traces

For clarity, the existing enumeration needs no fairness assumption. Encode a finite command word by continuation pointers:

$$\begin{aligned} e([]) &= 0, \\ e(\mathsf{S} :: v) &= 1 + p(0, e(v)), \\ e(\mathsf{MP}(x) :: v) &= 1 + p(1, p(x, e(v))), \\ e(\mathsf{AX}(c, x, y) :: v) &= 1 + p(c + 2, p(y, p(x, e(v)))). \end{aligned}$$

At a fetch boundary (n, l) the scheduler decodes n and increments n when $l = 0$; otherwise it decodes $l - 1$ and preserves n. Its residual pointer is precisely $e(v)$ on the displayed words. For any positive l, that residual is at most $l - 1$, since each Cantor projection is at most its input. Consequently each arbitrary candidate finishes after a bounded number of commands or halts. Fresh candidates enumerate $2, 3, 4, \dots$ from startup. Since W_d begins with AX, $e(W_d) - 1 \ge 3$, so its exact candidate is eventually

reached or an earlier command has already halted. Bounded induction through the preceding candidates is a PA proof of this assertion. Intermediate stores may be arbitrary: the universal property of W_d is exactly what is needed.

The source binding changes ordinal addresses explicitly:

	upstream PA196	PA195	PA194
halt site	10	10	10
fetch entry	25	25	25
AX entry	50	50	50
MP entry	3693	3581	3554
shared return	3799	3687	3660
Shift entry	3800	3688	3661
total primitive sites	3812	3700	3673

PA194 has 3504 AX sites and the same 169 exterior sites. Deleting one `cons` macro removes 27 primitive sites: 11 increments, 8 decrements, and 8 jumps. The independent ordinal control-flow checks bind every successor, not only the flattened token sequence, to the archived separate196 source. Thus the arithmetic-macro composition and the exact command transfer proof have the ten-entry AX effect above, with the displayed new MP, return, and Shift addresses. The fetch convention and original Shift/MP implementations remain identical under this ordinal map.

The final source program has 1457 increment, 1107 decrement, 1108 jump, and one halt primitive sites, giving $1457 + 2 \cdot 1107 + 1108 + 1 = 4780$ symbolic branch contracts. Both the unfolded and folded tables pass all of them. These quantify over the entire register file and an arbitrary affine unary gap; they are not 4780 concrete numerical examples. The maximum source path length is 38, below header constant 48.

Each symbolic contract consists of literal table rows, identical-tape reexpressions, self-loop traversals of runs, and the inherited register excursions. Induction on a run length or finite register file proves each excursion's trace. The register-box rows are preserved by the folds, and each accepted primitive trace has strictly positive physical length and its exact stated endpoint. Finite tapes, frame encodings, trace concatenation, and these inductions are arithmetizable in PA. The candidate-specific finite row equalities and completed symbolic traces therefore give a PA-provable positive simulation uniformly in all coded frames. The checked 58154-step blank startup, including every nonzero cell, ends at the clean fetch frame $(n, l) = (2, 0)$ and store $(0, 0, 0, 0)$. Its fixed finite trace is also PA-provable.

5.5 Sound stores and the two implications

To reflect a possible physical halt, annotate both formula registers t, h and all canonical stack entries with PA proofs of their translations T. The stack requires care: $p(0, 0) = 0$, so it is not a free list encoding. For $w > 0$, $\mathrm{tl}(w) < w$; define $\delta(w)$ as the least $k \leq w$ for which $\mathrm{tl}^k(w) = 0$. The annotation has exactly $\delta(w)$ entries, certifying the successive heads. A push prepends a proof except when both the code and stack are zero; then the canonical stack remains empty. Popping zero returns zero with its truth proof. The identities and compiler above also cover these cases, including the alias $I(0, 0) = 0$.

AX preserves this invariant by the PA suppliers already specified, including its unconditional separate Γ even on a failed guard. A successful MP combines the proofs of t and $h = I(t, v)$; its pushed zero receives a truth proof. A failed MP returns truth. A nonhalting Shift transfers the corresponding annotations. If Shift halts, $t + h = 1$ implies $t = 1$ or $h = 1$, so one annotation and the fixed PA proof of $\neg T(1)$ produce a PA refutation. Each update is primitive recursive and PA verifies its invariant.

Work now in PA under Con(PA). No annotated command can halt. The exact source command contract therefore supplies a positive execution to the next annotated fetch boundary. Given a proposed physical

halting time T, primitive recursion through $T+1$ command boundaries supplies more than T primitive source steps, and concatenating the positive primitive contracts supplies a physical execution longer than T. All these are bounded coded iterations with explicit finite witnesses. Determinism contradicts the proposed halt: a partial run cannot pass through the halt label and later reach a nonhalt frame. An alleged halt inside startup contradicts its fixed checked trace. Thus PA proves $H_{194} \to \neg\,\mathrm{Con(PA)}$. This argument needs neither inverse decoding of an arbitrary tape nor a least-time search for a future frame.

Conversely, a PA refutation gives a $\mathcal{C}_s$ refutation by (1) and Lemma 5.2. Its compiled W_d halts from every store. The bounded scheduler argument reaches that word or halts earlier; the arithmetic-macro and positive physical trace contracts then give a finite blank-tape halt of the literal table. All proof-producing operations and correctness inductions just used are PA-verifiable, so $\mathrm{PA} \vdash \neg\,\mathrm{Con(PA)} \to H_{194}$. The two implications prove Theorem 5.1.

6 Why this reduction succeeded

The successful step combined two established resources: the shorter separate190 emitter and the new PA195-style search for simultaneous physical folds. The exact separate190 source previously compiled to 196 controls. Its new association exposes eight compatible pairs, even though the raw AX allocation still has 68 controls. Optimizing only the number of raw grammar nodes would not distinguish the successful construction from several larger ones.

We also obtained a different, fully checked 195-state diagonal layout with 67 readers and six folds, and a separate195 layout with the same 67/6 count. These demonstrate that raw node count and final state count can move in opposite directions. The final 194-state result uses 68/8 instead of 67/6.

The exploratory work tested further local quotients, helper rewrites, core rotations, shorter scanners, alternate negation routines, parameter permutations, and logically equivalent placements of the diagonal clause. These searches informed the final choice; their negative outcomes are restricted to their specified families and verification procedures. One tempting 194-state matching from the shorter-scanner search failed a full source-frame check with an exact six-step symbolic cycle. Counting individually eligible edges was therefore insufficient. The final candidate is accepted only after the simultaneous local, complete source, full symbolic runtime, and blank-start checks all succeed.

From the ZFC work, the useful transferable ideas are predecessor history, shared helper paths, source reassociation, and exact positive simulation. Its consistency target itself is different and cannot be transferred by renaming a machine. From ConciseIPRA, the relevant methodological idea is to derive the interfaces a proof translation actually uses and to exploit syntactic freshness. Its Python proof-checker size is a different metric, and replacing full PA by iPRA would change the target consistency statement. The present finite-comprehension interpretation still uses induction for an arbitrary arithmetic formula.

No minimality theorem for 194 states is asserted. The result is the explicit, verified strict improvement requested here.

7 Reproduction and package contents

The reconstruction is deterministic and requires only Python 3 and its standard library. It does not rerun the heuristic search. From the package directory, run

```
python3 -B pa194/build_verify.py
python3 -B verification/independent_verify.py
python3 -B mathematics/check_emitter_and_words.py
```

Run without Python's `-O` flag. The scripts reconstruct their objects and compare them with the delivered tables and certificates; they do not accept a success label in a saved JSON file as a proof. The candidate association, literal rows, native/dense bijection, full source, symbolic emitter identity, local blocks, and startup certificate are all included. The pinned reference packet retains its licenses and original attribution.

To build this article, run twice from `report/`:

```
pdflatex -interaction=nonstopmode -halt-on-error pa194_conpa.tex
```

Alternatively use `latexmk -pdf`. No downloaded bibliography or external figure is required. `SHA256SUMS` binds every delivered file other than that manifest itself.

References

[1] Supplied PA195 construction and written internal-PA calibration, `docs/incoming/pa195_conpa_2026-11-13.zip`, ProveIt snapshot `5be8b9236df6a6d308195e62a906a173dc70d00a`. `https://github.com/VladimirReshetnikov/ProveIt/blob/5be8b9236df6a6d308195e62a906a173dc70d00a/docs/incoming/pa195_conpa_2026-11-13.zip`. An unchanged copy is included under `reference/pa195_conpa/`.

[2] Supplied diagonal196 and separate196 constructions, `pa_consistency_comparison_and_new196_2026-11-13.zip`, same snapshot. `https://github.com/VladimirReshetnikov/ProveIt/blob/5be8b9236df6a6d308195e62a906a173dc70d00a/docs/incoming/pa_consistency_comparison_and_new196_2026-11-13.zip`.

[3] James Chang, PA196 and PA209 developments, imported under ProveIt's PA research directory. The inherited PA196 source is pinned by its archived provenance to upstream revision `407cdc356f6b586635e71ac92be394255f4bfea4`. `https://github.com/VladimirReshetnikov/ProveIt/tree/5be8b9236df6a6d308195e62a906a173dc70d00a/Computability/BusyBeaver/Research/PA/jamesyc-upstream`.

[4] Supplied cross-archive compositions, `pa_consistency_cross_archive.zip`, same snapshot, Git blob `303d864d7434c0e2a50a3b4d6e5229578bf9c227`.

[5] Supplied reduction audit, `pa_consistency_reduction_audit_2026-11-14.zip`, same snapshot, Git blob `3f37e13d347b8e4929199a631793d97a25dc0f07`.

[6] Supplied negation and emitter research, `pa_consistency_research_2026-11-14.zip`, same snapshot, Git blob `af1db044bb77bc29d960e18895fcdecd69edb0d8`.

A Complete transition table

In the following paired-column presentation, an action wDq' means write w, move D, enter q'. The two action columns are for reading zero and one. The halt label 194 has no row. The machine starts at state 0 on an all-zero tape.

q	Read 0	Read 1	q	Read 0	Read 1
0	1R1	1R61	97	0R106	1R8
1	1R82	1R1	98	0R91	1R107
2	0L2	1L3	99	0L131	0L85
3	0L4	1L3	100	0R108	1R94
4	0L6	1L3	101	0L109	0L2
5	1R7	0L5	102	0R38	1R68
6	0L9	1L3	103	0R59	1R110
7	0R7	0R10	104	0R59	1R111
8	0R14	1R13	105	0R66	0L16
9	0L15	1L3	106	0R14	1R110
10	0L16	0R10	107	0R21	1R112
11	0L26	1R18	108	0R113	1L20
12	0R19	1R20	109	0R30	1R65
13	0L79	1L31	110	0R38	1L108
14	0L11	0R22	111	0R121	0L16
15	0L23	1L3	112	0R114	0L16
16	1L2	0L2	113	0R24	0R47
17	0R24	0R32	114	0R48	0R32
18	0R84	0L95	115	0R70	1R86
19	0R28	1R29	116	0R19	1R118
20	0R30	1R31	117	0R126	1R28
21	0R24	0R57	118	0R122	1R107
22	0R33	1R22	119	0R94	1R86
23	0L34	1L3	120	0R83	1R98
24	0L35	0R1	121	0R87	1R87
25	0R36	1R25	122	0R17	1L109
26	0R122	1R41	123	0R70	1R101
27	0R38	0L39	124	0R96	1R108
28	0R40	1R41	125	1L39	1L99
29	0R42	1R43	126	0R80	1R78
30	0R44	1R45	127	0R120	1R133
31	0R21	1R46	128	0R124	1R123
32	0R47	1R32	129	0R123	1R128
33	0R32	1R33	130	0R44	1R90
34	0L69	1L3	131	0L5	0L131
35	1R35	1L2	132	0R127	1R59
36	0R49	1R36	133	0R11	1R37
37	0R28	1R20	134	0R135	1R193
38	0L11	0R36	135	0R136	1R140
39	0L5	1L51	136	0R12	1R137
40	0R24	0R25	137	0R138	1R58
41	0R13	0L16	138	0R86	1R139
42	0R53	1R54	139	0R78	1L164
43	0R55	1R56	140	0R141	1R188
44	0R24	0R52	141	0R142	1R187
45	0R58	1R59	142	0R64	1R143
46	0R60	0L16	143	0R144	1R176

q	Read 0	Read 1	q	Read 0	Read 1
47	0R25	1R47	144	0R12	1R145
48	0L35	0R61	145	0R146	1R175
49	0R16	1R49	146	0R86	1R147
50	0R63	1R64	147	0R148	1R169
51	0L39	1L51	148	0R78	1R149
52	0R22	1R52	149	0R150	1R162
53	0L11	0R49	150	0R126	1R151
54	0R58	0L16	151	0R152	1R157
55	0R65	1R66	152	0R157	1R154
56	0R59	1R67	153	0L183	1R78
57	0R52	1R57	154	0R155	1R156
58	0R48	0R25	155	0R116	1R153
59	0L28	1L68	156	0R139	1R136
60	0L102	1L30	157	0L145	1R153
61	0R0	1R61	158	0R159	1R160
62	1R125	1L51	159	0R160	1R137
63	0R70	1R194	160	0R19	1R161
64	0L126	1R31	161	0R58	1L190
65	0R40	1R60	162	0R181	1R166
66	0L79	1R71	163	0R12	1R164
67	0R53	1L72	164	0R153	1R87
68	0R73	1R74	165	0R137	1R136
69	0R69	1R75	166	0R164	1R167
70	0R76	1R77	167	0R78	1R168
71	0R79	1L110	168	0L156	1L175
72	0L98	1R54	169	0R170	1R174
73	0R48	0R33	170	0R163	1R171
74	0R81	0L41	171	0R161	1R173
75	0R50	1R132	172	0R116	1R138
76	0R130	1R21	173	0R156	1R58
77	0R101	1R90	174	0R155	1R171
78	0R92	1R93	175	0R158	1R172
79	0R73	1R72	176	0R177	1R184
80	0R91	1R119	177	0R178	1R166
81	0R95	1R96	178	0R179	1R183
82	0R62	0R1	179	0R180	1R182
83	0R27	1R97	180	0R58	1R181
84	0R89	1R88	181	0L169	1R165
85	1L99	1L85	182	0L185	1L176
86	0R19	1R100	183	0R182	1R168
87	0R58	1R58	184	0R185	1R116
88	0R117	1R115	185	0R186	1R186
89	0L26	1R134	186	0L154	1R172
90	0R95	0L131	187	0R168	1R78
91	0R102	1R103	188	0R189	1R191
92	0R67	1R56	189	0R159	1R190
93	0R55	1R104	190	0R186	1R187
94	0R21	1R105	191	0R164	1R192
95	0R21	0L10	192	0R168	1R122
96	0R59	1R41	193	0R129	0L131

B Archive-by-archive comparison

The following 27 entries carry forward the detailed, classified comparison in the supplied PA195 report. Their contribution and evidence status refer to those archived designs, not to the new PA194 table. The four further entries complete the current 31-archive scope. Exact current archive identities and every recognized table member are bound by the fresh census.

Archive	Contribution and status
U208_ConPA_2026-10-29.zip	Complete U208, original class A: identify old native roles 172 and 211, with helper 217 (or 214 under alternate native labeling). The local simulation is a finite all-tape argument; full PA calibration is inherited and written, not newly kernel exported.
U208_ConPA_Conditional_Context_2026-10-29.zip	Same conditional-history reduction; primary and alternate presentations give the two original U 208 graph classes. Four local blocks have lengths 7,1,3,1.
U208_ConPA_Entry_Context_2026-10-29.zip	Re-expresses the same reduction by entry guards, with primary and compatibility alternatives. Its saving is not additive with the previous packets.
U208_ConPA_History_2026-10-29.zip	Historical-label presentation of the same two classes and same one-state deletion; includes a failed further reduction.
u208_research_2026-10-29.zip	Both original U 208 classes with symbolic runtime and startup evidence; no further state saving.
pa208_research_2026-10-29.zip	Eighteen additional pairwise nonisomorphic U 208 helper variants, each with a local halting-equivalence argument. Expands accepted U 208 corpus to 20 classes; no accepted 207.
Below_U208_Research_2026-10-29.zip	Source-boundary guards, displaced-head quotient lemma, closed-loop pruning, and broader negative local searches. Pruning 3909 to 3904 source occurrences increases the shared-tree cost. No accepted 207.
ConPA_below208_research_2026-10-29.zip	Two-object obstruction to collapsing source/output cover roles, exhaustive restricted axiom-parameter permutation search, two-row and register-invariant searches. No accepted 207.
pa_consistency_research_2026-10-29.zip	Proves zero-guarded source specialization 3909 to 3904 instructions and preserves the ordinal control graph; no smaller physical table.
u208_comparison_research_2026-10-29.zip	Exact comparison, source/encoding research and a fully specified rejected 207 frame witness. Weak/exact schema equivalence is separate from the still-open runtime/calibration replacement.
pa_unary_power_cover.zip	A unary-matrix power-cover theory and written mutual consistency calibration with full PA. Supplies numerical proof-search interface and logical separations; no new literal TM.
pa_tm_research_2026-11-13.zip	New alternative 196 with head order (1,2,3,5,6,4), unchanged 3812-site CFG, maximum source depth 36 rather than 38. Physical Lean baseline rebuild is supplied. R6-pruned 198 passes runtime but lacks calibrated blank startup.
pa_tm_extended_research_2026-11-13.zip	Four distinct one-row closing-CD variants remain196, with constrained symbolic runtime and unchanged blank startup. Joint source invariants and a generalized register executor strengthen search.
conpa_tm_comparison_and_search.zip	Accepted AD-register-swap196. Two-rotation 195 is rejected by 93 primitive-contract failures. Source geometry changes cannot reuse old contextual folds automatically.
pa196_research_2026-11-13.zip	Exact 157-node representation of the original 3812-site source, down from159; seed stays at source path100000. Claimed prospective 195 allocation loses a required decoder helper. Native HA/full-induction theory is a separate research direction.

Archive	Contribution and status
pa_consistency_research.zip	New 158-node source and an accepted 196 regrouping; the eighth scanner zero consumes the apparent one-state saving. Its seven-zero 195 version misroots on a concrete source frame.
pa_consistency_reduction.zip	Rebracketed196; H-only Shift 196 with 3811 sites and 57922-step startup; integrated witness gate has 3745 sites but best physical table203. Two optimistic 195 allocations fail, one by an exact short cycle on a source frame.
pa_tm_comparison_and_reductions.zip	Two accepted 196 rotations. Equivalent gate reordering shortens AX 201 to 199 macro items and source 3812 to3773, but best physical compilation198. Finite free-algebra/logic certificates and restricted grammar searches are retained.
pa_consistency_research_2026-11-13.zip	Diagonal vacuity B→U(B,B), exploiting formula-code freshness; AX 201 to191, source 3812 to3700. Initial complete implementation199. Gives detailed written reverse PA proof translation; no new full Lean endpoint.
pa_consistency_research_2026-11-13(2).zip	New twelve-rotation 196 and diagonal198. Also a command calculus halting at successful MP deriving code1; no new binary table for that command change.
conpa_tm_comparison_and_reduction_search.zip	Two experimental 196 MP commands (result in H, omit zero push), with mathematical command arguments but only bounded physical frame evidence. Two rejected Shift layouts actually halt from blank after 43563 steps.
conpa_tm_study_2026-11-13.zip	Combined 196 implements H-only Shift and MP-in-H, with 4928 uniform contracts; an additional rebracketed Combined 196 has full runtime and startup confirmed in the PA195 audit; direct-negation 197 has 3556 sites. A 157-node source receives a larger 207-state realization.
pa_consistency_comparison_2026_11_13.zip	First quantifier head split into modal B plus explosion, with finite Hilbert certificates in both directions. Countermodels obstruct deleting either membership rewrite head or replacing the first quantifier head by reflexivity. Twelve source/table combinations retain separate command/calibration obligations.
pa_consistency_comparison_and_new196_2026-11-13.zip	Three accepted diagonal-calculus 196 layouts: diagonal 196 (191 AX items,3700 sites), separate 196 (190 items,3673 sites), scanner_best 196 (same 3673 sites, improved address layout). Also a runtime/startup-checked scanner 196 on original source and four auxiliary diagonal closing-CD variants with differential runtime proofs, trit invariants, and exact startup. Short-scanner 195 and U208/PA 196 hybrid startup attempts fail stated obligations.
pa_below196_research.zip	Eight equivalent gate spellings, orientation involutions, exact freshness obstruction R(5)=9, smaller pairing macro that increases DAG cost, H-only guard specialization, and broader finite physical searches. No new accepted smaller table.
pa_tm_research_2026-11-13(2).zip	New zerojump 196 and an auxiliary CD3 parent (fully replayed in the PA195 audit): native27/1 changes 0 L40 to1 R21 under the source26 R6=0 invariant; startup58402. Deleting the resulting prospective helper gives a 195 table with wrong source successor and R6 increment, witnessed also on the actual blank execution.
PA_consistency_TM_research_2026-11-13.zip	New source-only zero-run-five witness (3812 sites,159 internal nodes,depth37), combined-negation/MP 3573-site 157-node witness, and parameter-free residual MP. Fixed/protected physical allocations remain rejected; many bounded quotient searches are negative.

Archive	Contribution and status
`pa195_conpa_2026-11-13.zip`	The supplied 195-state baseline: 191 AX items, 3,700 source sites, 4,815 contracts, seven compatible folds. Full ordinary internal-PA calibration, complete startup, and independent audit. This is the comparison threshold for the present task.
`pa_consistency_cross_archive.zip`	Four 196-state combinations of diagonal/separate emitters with H-only Shift and MP in H, with optional invariant-based zero jump. Source counts 3,676/3,649; complete runtime and startup evidence. Ten-entry selectors for its changed commands must not be confused with the original-command selectors used by PA194.
`pa_consistency_reduction_audit_2026-11-14.zip`	Diagonal/separate emitters plus MP in H remain 196 states (3,677/3,650 source sites); adding shorter negation gives 197. Larger parameter and helper searches; no supported smaller table.
`pa_consistency_research_2026-11-14.zip`	An 18-instruction negation macro with a zero/positive scratch invariant checked on 245,112 abstract pairs. The full separate-emitter implementation has 197 states, 3,441 source sites and 4,468 symbolic contracts. The source saving loses physical sharing.